SUSTAINABILITY LAB · PROF. NIPUN BATRA · IIT GANDHINAGAR

AI and sensing for a sustainable future.

We build sensors, datasets, benchmarks and machine-learning models for clean air, Earth observation, health and energy — and release them openly.

CORE METHODS

Vision-language models
ThermEval · thermal-image reasoning

LLMs and agents
VayuBench · VayuChat · VayuBuddy

Computer vision
SentinelKilnDB · Space to Policy · PlantDoc

Generative models
FlowSat · diffusion and flow matching

Gaussian processes
Scalable GPs · Deep GPs · sensor placement

Active learning
Sensor placement · NILM · SpiroActive

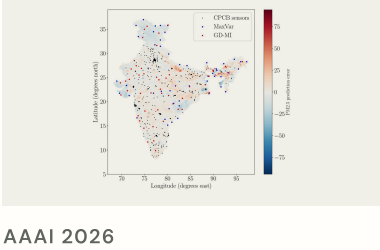
Self-supervised learning
Signal descriptors for time series

Sensing systems
Thermal, acoustic and edge sensing · JoulesEye · SpiroMask · ApneaEye

01

Clean Air

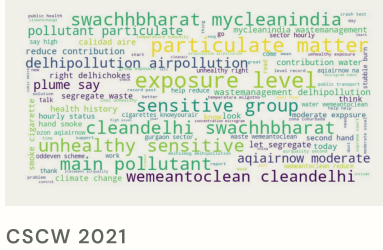
Most Indian cities have few air monitors. We build low-cost sensing, models that fill the gaps, and study how the public talks about air.



AAAI 2026

Sensor placement

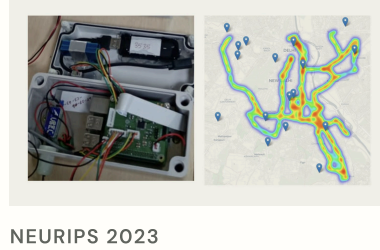
Gradient-based mutual information picks where to put the next monitor, at continental scale.



CSCW 2021

Vartalaap

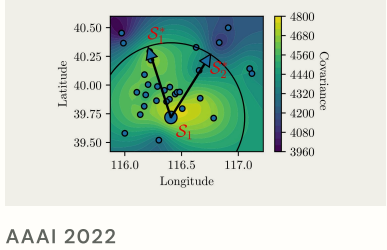
What drives India's air-quality talk on Twitter: politics, pollution or pseudo-science?



NEURIPS 2023

AirDelhi

A fine-grained mobile PM2.5 dataset from sensors mounted on Delhi's public buses.



AAAI 2022

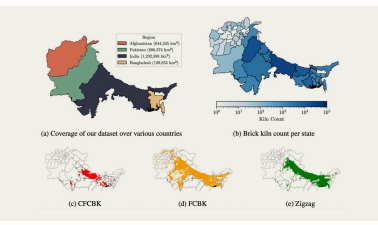
Scalable GP inference

Gaussian processes that estimate air quality between monitors, accurately and at scale.

02

Earth Observation

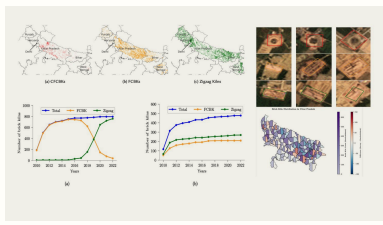
Polluters such as brick kilns are spread over areas too large to survey. We find them from space and link them to policy.



NEURIPS 2025

SentinelKilnDB

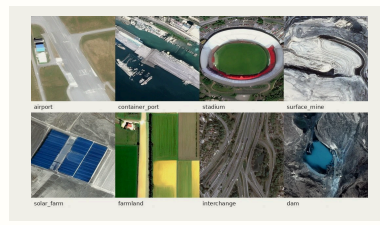
A hand-validated benchmark for oriented brick-kiln detection across South Asia.



ACM JCSS 2025

Space to Policy

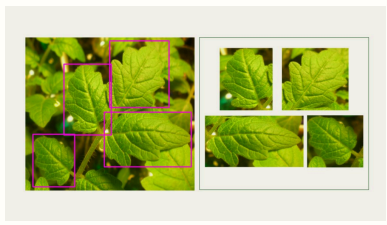
Satellite kiln detections checked automatically against siting regulations.



BMVC 2026

FlowSat

Diffusion transformers generate realistic satellite images from acquisition metadata.



CODS 2020

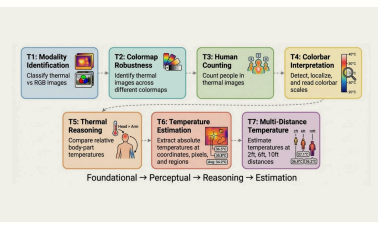
PlantDoc

An open dataset for detecting crop disease from photos taken in the field.

03

Health Sensing

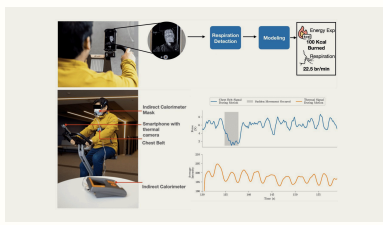
Measuring lung function or calories burned usually needs a clinic. We turn cameras, masks and wearables into instruments.



KDD 2026

ThermEval

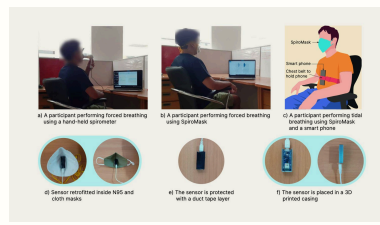
A 7-task benchmark showing where vision-language models fail on thermal imagery.



IMWUT 2024

JoulesEye

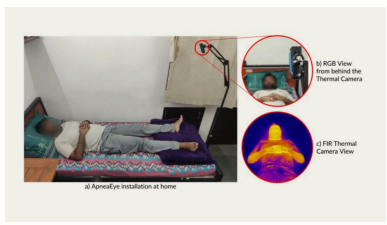
A thermal camera estimates energy expenditure and breathing during exercise.



ACM TCH 2023

SpiroMask

An N95 mask with a microphone measures lung function without a spirometer.



PATENT FILED 2025

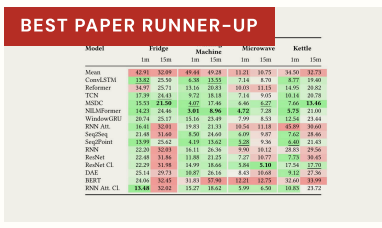
ApneaEye

Contact-free thermal-camera screening for sleep apnea and breathing disorders.

04

Energy

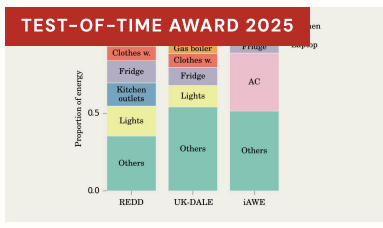
An electricity bill says how much a home used, not what used it. Since 2014 we have built the tools the field uses to answer that.



BUILDSYS 2026

NILMBench2026

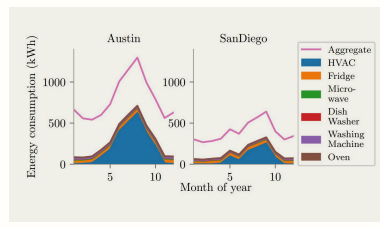
A reproducible benchmark of 16 disaggregation models; led by second-year undergraduates.



E-ENERGY 2014

NILMTK

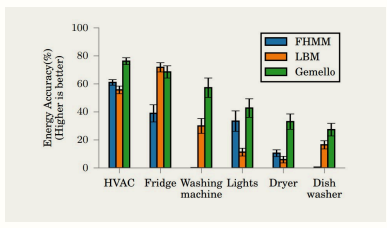
The open-source toolkit that made energy-disaggregation research reproducible.



AAAI 2018

Cross-region transfer

Learns energy breakdown in one city and transfers it to regions with little data.



KDD 2016

Gemello

A detailed appliance breakdown from nothing more than a monthly electricity bill.

INFRASTRUCTURE

Compute for every student.

Shared GPU servers on campus, available to every lab member.

4× A100 · 80 GB

2× RTX A5000 · 24 GB

2× RTX A4000 · 16 GB

960 GB RAM

112 CPU cores



PRINCIPAL INVESTIGATOR

Prof. Nipun Batra

Associate Professor, Computer Science & Engineering, IIT Gandhinagar

Nipun completed his PhD at IIIT Delhi and postdoctoral research at the University of Virginia. The lab works where machine learning meets sensing for computational sustainability — with an emphasis on open, reproducible work that gets deployed.

01 Recognition

LAB & PI

2026	**Best Paper Runner-Up**	NILMBench2026 · ACM BuildSys
2025	**Test of Time Award**	NILMTK · ACM e-Energy
2025	**ACM SIGEnergy Rising Star**	Nipun Batra
2025	**Excellence in Teaching Award**	Nipun Batra · IIT Gandhinagar
2023	**Young Alumni Award**	Nipun Batra · IIIT Delhi
2015	**Best Graduate Presentation**	Nipun Batra · ACM SenSys

STUDENTS

2026	**Three IndiaAI MTech Fellowships**	Balbir Prasad, Saaransh Shandilya, Parv Thacker
2025	**Outstanding Research Commendation**	Rishiraj Adhikary · IITGN Convocation
2024	**Microsoft Research India PhD Award**	Zeel B Patel · one of ten in India
2024	**Heidelberg Laureate Forum**	Rishiraj Adhikary · one of 200 worldwide
2023	**Gold Medal for Outstanding Research**	Ankita Jain · MTech
2022	**Fulbright-Nehru Doctoral Fellowship**	Rishiraj Adhikary

02 Selected papers

PHYS. REV. D 2025	Constraining the 3HDM Parameter Space using Active Learning
BUILDSYS 2021	Neural Network Approaches and Dataset Parser for NILM Toolkit
WWW 2019	A Tree-Structured Neural Network Model for Household Energy Breakdown
BUILDSYS 2019	Towards reproducible state-of-the-art energy disaggregation
BUILDSYS 2022	"I do not know": Quantifying Uncertainty in Neural Network Based Approaches for Non-Intrusive Load Monitoring
BUILDSYS 2020	EdgeNILM: towards NILM on edge devices
CIKM 2019	Active Collaborative Sensing for Energy Breakdown
AAAI 2017	Matrix Factorisation for Scalable Energy Breakdown

03 Alumni

2 PhD

13 MTech

Dr. Zeel B Patel — PhD 2026
Postdoc at Microsoft Research India

Dr. Rishiraj Adhikary — PhD 2025
Postdoc at University of Cambridge (with Prof. Cecilia Mascolo)

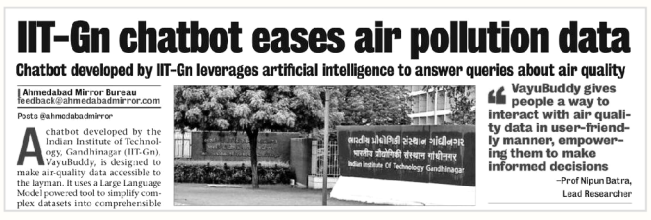
ALUMNI WENT ON TO

Google DeepMind · Microsoft Research · Carnegie Mellon University · Caltech · UC San Diego · University of Illinois Urbana-Champaign · UMass Amherst · Oracle

SUPPORTED BY

Google · Microsoft · NVIDIA · Cisco · Jio · SERB · Department of Science & Technology · STARS · Ministry of Education

04 In the press



VayuBuddy · Ahmedabad Mirror



SpiroMask · Ahmedabad Mirror



JoulesEye · The Indian Express

Join us.

PhD · Postdoc · Research associate · Internship

We look for people who like building things that work outside the lab. No prior sustainability background needed.

Openings

Lab website